



Evaluating sustainability of CO₂-mediated pyrolysis of lignocellulose

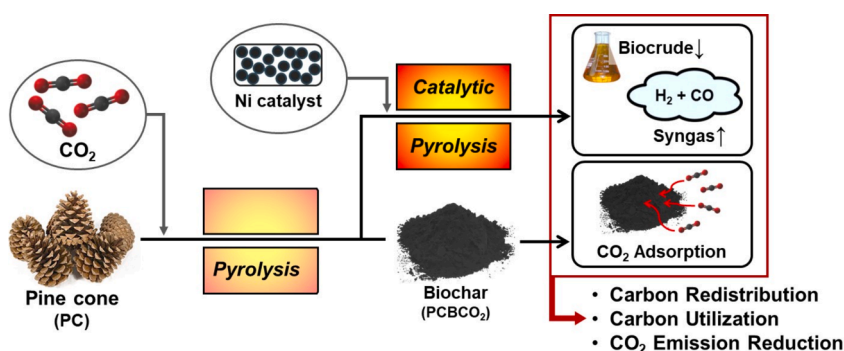
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HIGHLIGHTS

- Lignocellulosic biomass (pine cone, PC) was pyrolyzed using CO₂ as a reactive gas.
- CO₂ enhanced syngas production by interacting with volatile compounds from PC.
- Catalytic pyrolysis with CO₂ improved the conversion of PC into syngas (CO).
- Increased porosity of biochar (PCBCO₂) improved its CO₂ adsorption capacity.
- Sustainability of proposed pyrolysis was evaluated by estimating net CO₂ emissions.

GRAPHICAL ABSTRACT



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ABSTRACT

Despite the growing interest in biomass as a carbon-neutral resource, technical challenges have limited its comprehensive utilization. Pyrolysis has emerged as a promising method for reducing the carbon footprint by more effectively valorizing carbon in biomass. This study investigated the use of carbon dioxide (CO₂) in the pyrolysis of pine cone (PC), a lignocellulosic biomass. Thermogravimetric analysis confirmed that lignin was the primary component of the PC. Characterization and quantification of the three pyrolytic products (syngas, biocrude, and biochar) revealed that CO₂ enhanced CO production and the surface area of the biochar, thereby improving its CO₂ adsorption capacity. Additional heat and a Ni catalyst further amplified CO₂'s functionality. The sustainability of the proposed pyrolysis system was evaluated by calculating energy requirements of the pyrolysis processes and the net CO₂ emissions. Catalytic pyrolysis under CO₂ was the most effective, achieving a reduction of 3.34 g of CO₂ per gram of PC.

1. Introduction

Energy availability is a key factor in determining quality of life, as economic activities are closely linked to energy consumption (Banna

et al., 2023; Tayyab Ayaz et al., 2024). Fossil fuels have been extensively used as the primary energy source due to their high energy density. However, this extensive use has resulted in significant environmental problems, especially global warming. Anthropogenic carbon emissions

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from fossil fuel combustion reached approximately 36.3×10^9 tons CO₂-equivalent in 2022, marking an 18.6-fold increase over the past 12 decades (Friedlingstein et al., 2023; Ghorbani et al., 2024). This amount far exceeds Earth's yearly natural CO₂ assimilation capacity, which is estimated to be 24.2×10^9 tons CO₂-equivalent (Friedlingstein et al., 2023). To address this carbon imbalance, biomass utilization has gained attention as an alternative to fossil fuels, given its inherent carbon neutrality (Poornima et al., 2024). However, biomass may not be the most suitable choice for primary energy recovery because of its lower calorific values than fossil-based fuels (Verevkin et al., 2022). Also, complete combustion of biomass cannot be readily achieved due to its poor diffusion rate of oxidant (air) (Williams et al., 2012). The technical challenges in achieving an optimal stoichiometric ratio create unfavorable conditions that result in the production of air pollutants (Williams et al., 2012). Therefore, it is crucial to investigate strategic methods of transforming biomass into combustible fuels.

Given this premise, the production of biofuels such as biodiesel and bioethanol has gained prominence because of their compatibility with existing infrastructure, such as internal combustion engines (Barua et al., 2023). In addition, their blends with commercial fuels such as diesel and gasoline have been mandated under stringent regulations such as the Renewable Portfolio Standard and Renewable Fuel Standard (Morone et al., 2020). Despite their increasing application in the transportation sector, the processes for converting source materials into biodiesel and bioethanol still face technical challenges in effectively extracting carbon (Xu et al., 2019). Biodiesel and bioethanol are typically synthesized from lipids and carbohydrates in biomass, respectively, leaving unconverted carbon waste (Ashokkumar et al., 2017). An alternative method for extracting more carbon from biomass is offered by anaerobic digestion, because this process enables the conversion of biodegradable components such as proteins, lipids, and carbohydrates into biogas, particularly methane (CH₄) (Bharathiraja et al., 2018). However, uncontrolled microbial metabolism, including cell growth and respiration, leads to considerable carbon loss in the form of sludge and CO₂ (Hu et al., 2015). Moreover, the biologically recalcitrant components of biomass, particularly lignin (15–40 wt%) (Kataeva et al., 2013), are considered non-usable carbon. Therefore, it is crucial to explore biomass refinement techniques to maximize carbon utilization.

In this context, thermochemical processing, such as pyrolysis, shows potential for more comprehensive carbon valorization of biomass. Pyrolysis distributes the carbon content of the biomass into three products: syngas, biocrude, and biochar (Bing et al., 2024). The composition of these products can be adjusted by manipulating various process parameters, including the feedstock type, reaction medium, temperature, heating rate, and catalyst type (Shafizadeh et al., 2023). Biomass pyrolysis is often used for the production of biocrude (Guedes et al., 2018). However, biocrude is composed of a diverse array of chemical components with varying functional groups and carbon numbers (Staš et al., 2020), resulting from the thermal cracking of polymeric components in biomass. Despite its potential as a liquid fuel, the complex composition of biocrude requires post-treatment such as upgrading and separation (Lahijani et al., 2022). The additional post-treatment process steps significantly affect the economic feasibility of biocrude applications (Yildiz et al., 2016). To overcome these challenges, it may be advantageous to convert biomass into gaseous products, particularly syngas (H₂ and CO mixtures), owing to its simpler control and direct fuel application (Wen et al., 2022).

Based on these considerations, this study aimed to develop a pyrolysis system to improve the production of combustible gaseous fuels. To achieve this, a softwood logging residue, pine cones (PC), was used as a representative lignocellulosic biomass (Sharma et al., 2023). The rapid growth rate and global abundance of softwood have made it the preferred material for wood-based products (Vepakomma et al., 2011). However, 20–40 wt% of softwood is left as residues (Greiner et al., 2019). During pyrolysis, CO₂ was introduced to further improve the carbon distribution among the pyrogenic products. CO₂ is often used as

an oxidative agent in gasification processes, which typically occur at temperatures ≥ 700 °C (Vamvuka and Sfakiotakis, 2015). However, this study primarily focused on exploring the functional reactivity of CO₂ at reaction temperatures ≤ 700 °C. This approach could help optimize syngas production (Di Blasi, 2009). Although various transition metals are adopted due to their capabilities to dissociate the chemical bonds, nickel (Ni) was specifically chosen for its superior cost-effectiveness (Luo et al., 2022; Wu et al., 2013). Qualitative and quantitative analyses of all pyrolytic products obtained under CO₂ were compared with those obtained under N₂. Lastly, the energy consumptions and the net CO₂ emissions were calculated to assess the practical feasibility of the proposed pyrolysis processes.

2. Experimental section

2.1. Sample preparation and chemical reagents

PC was obtained from a local wood farm in South Korea. The sample was dried in a convection oven at 90 °C for 24 h to remove moisture. The dried samples were pulverized and sieved to obtain a uniform particle size (less than 0.1 mm). Aluminum oxide (Al₂O₃, Lot # A0119R11), dichloromethane (CH₂Cl₂, 99.8 %, Lot # D0214YD1), and nickel (II) nitrate hexahydrate (Ni(NO₃)₂·6H₂O, 97.0 %, Lot No. N0024XE1), and n-octyl alcohol (C₈H₁₈O, 98 %, Lot # O0005XK1) were purchased from Daejung Chemicals Co., Korea. Hemicellulose (Lot # F2RXN-KH) was purchased from Tokyo Chemicals Co., Japan. Cellulose (Lot # WXBC0699V) and lignin (Lot # MKCL2371) were purchased from Sigma-Aldrich Co., USA. An inorganic calibration mixture (standard solution IV, Lot # HC60659055) for inductively coupled plasma (ICP) analysis was purchased from Merck Co. (Germany). Ultra-high-purity (≥ 99.9 %) grade N₂ and CO₂, and zero-grade air were purchased from Green Gas Co., Korea. A gas-calibration mixture containing N₂, CO₂, H₂, CO, CH₄, C₂H₆, C₂H₄, and C₂H₂ was obtained from RIGAS Co., Korea.

2.2. PC characterization

The thermal breakdown pattern of PC under N₂, CO₂, and air was investigated by thermogravimetric analysis (TGA, STA449 F5 Jupiter, NETZSCH, Germany). For each TGA run, 10.00 ± 0.02 mg of PC was loaded into an aluminum crucible in the analyzer. The flow rates of N₂, CO₂, or air were set at 80 mL min⁻¹, with an additional flow of N₂ at 20 mL min⁻¹ as a balance protecting gas. The temperature was programmed from 40 to 900 °C at 10 °C min⁻¹. Elemental analysis using EA3100, EuroVector, Italy, revealed that PC was composed of 46.7 wt% C, 5.9 wt % H, 44.9 wt% O, and 0.4 wt% N.

2.3. Pyrolysis setup

Three experimental setups were employed for pyrolysis tests: conventional single-step, multi-step, and catalytic pyrolysis. A schematic representation of these three pyrolysis setups is shown in Figure S1. For the conventional setup, a cylindrical heat block (PTF-804-2, LAB House, Korea) and a quartz tube reactor (2.2 cm inner diameter, 0.3 cm thickness, and 120 cm length) were used. The vacuum system was established using two fittings (SS-1610-6-4 and SS-16-UT-6; Swagelok, USA) at both ends of the cylindrical reactor. A sample of PC (5.00 ± 0.01 g) was placed at the center of the heat block. The flow rate of N₂ or CO₂ was maintained at 800 mL min⁻¹. The temperature program for the heat block was set to increase 100–700 °C at a steady rate of 10 °C min⁻¹. The experimental conditions for the multi-step pyrolysis process were same with those employed in conventional pyrolysis. However, an additional cylindrical heat block, operating isothermally at 600 °C, was installed adjacent to the existing heat block. For the catalytic pyrolysis, a Ni catalyst was placed at the center of the additional heat block while maintaining the same operational parameters as those used in the multi-step pyrolysis process.

The Ni catalyst was synthesized using an impregnation method reported previously (Lee et al., 2024). X-ray diffraction (XRD, mini-Flex600, Rigaku, Japan) was employed to evaluate the crystallinity of the synthesized catalysts following the analysis conditions described in a previous study (Valizadeh et al., 2021). The obtained XRD spectrum confirmed the presence of γ -Al₂O₃ and metallic Ni (Figure S2). ICP-optical emission spectrometry (ICP-OES, Optima 5300DV, PerkinElmer) revealed the Ni content in the catalyst was 4.8 wt%.

2.4. Pyrolytic product analysis

The PC pyrolysis produced both condensable (biocrude) and non-condensable (syngas) volatile chemical products. The condensable chemicals, were collected through a cold trap maintained at -40°C at the end of the pyrolysis reactor. The condensed chemicals were dissolved in and diluted with dichloromethane and characterized using gas chromatography-mass spectrometry (GC-MS, 8890-5977B, Agilent, USA). The GC-MS system was equipped with a capillary column (DB-5MS, 30 m $\times$ 0.25 mm $\times$ 0.25 μm). Non-condensable gases were measured online using a micro-gas chromatograph (990, Agilent, USA) equipped with a thermal conductivity detector and dual columns, a packed column (Molecular 5A) and a capillary column (Plot U). The detailed operating conditions for the GC-MS and micro-GC are summarized in Table S1. The concentrations of gaseous products, including syngas (H₂ and CO) and C₁₋₂ hydrocarbons (CH₄, C₂H₆, C₂H₄, and C₂H₂), obtained from the N₂ and CO₂ conditions were calculated using the following equations (1) and (2), respectively.

$$\text{Gasproduction}(\text{mmol}) = \frac{800\text{mL}}{V_N} \times \frac{V_g}{V_m} \times \frac{273\text{K}}{298\text{K}} \quad (1)$$

$$\text{Gasproduction}(\text{mmol}) = \frac{160\text{mL}}{V_N} \times \frac{V_g}{V_m} \times \frac{273\text{K}}{298\text{K}} \quad (2)$$

Where V_m is the molar volume (mL mmol^{-1}) at a standard temperature and pressure condition, V_N is the volumetric content of N₂ (vol.%), and V_g is the volumetric content of each gas (vol.%).

The solid residue (biochar) remaining after the pyrolysis tests was collected from the pyrolysis reactor. The surface characteristics of biochars (PCB) produced in N₂ and CO₂ environments at 700°C , including specific surface area, mean pore volume, mean pore size, and pore size distribution, were determined using a Brunauer-Emmett-Teller (BET)/Barrett-Joyner-Halenda (BJH) analyzer (BELSORP-mini II, MicroTrac, Japan). The PCB's surface morphologies were examined by scanning electron microscopy (SEM; EmCrafts, CUBE 2, Korea). The CO₂ adsorption capacity of the PCB's was evaluated using a TGA instrument, following a previously reported method (Lee et al., 2020). The biochars, with a loading of 50.00 ± 0.01 mg, were pretreated at 200°C under a N₂ flow (100 mL min^{-1}) for 0.5 h to remove impurities, including moisture and remaining volatiles, and then cooled down to reach 25°C . For the CO₂ adsorption test pretreatment, the N₂ gas was converted into CO₂ gas (80 mL min^{-1} with a balance N₂ gas (20 mL min^{-1})) while maintaining 1 h. The resulting CO₂ adsorption curves of the biochars were plotted

Table 1
Two fitting models used for CO₂ adsorption capacity.

Fitting model	Equation	Parameter	Ref.
Pseudo-first-order (PFO)	$q_t = q_e \times (1 - e^{-k_1 \times t})$	q_e : equilibrium adsorption capacity, $[\text{mmol g}^{-1}]$ q_t : adsorption capacity at time t , $[\text{mmol g}^{-1}]$ k_1 : PFO constant, $[\text{min}^{-1}]$ t : time, $[\text{min}]$	(Simonin, 2016)
pseudo-second-order (PSO)	$q_t = (k_2 \times q_e^2 \times t) \times (1 + k_2 \times q_e \times t)^{-1}$	k_2 : PSO constant, $[\text{g mmol}^{-1} \text{min}^{-1}]$	(Chen et al., 2019)

with two numerical models and the details for the kinetic evaluation results are listed in Table 1.

2.5. Evaluating sustainability of pyrolysis test

2.5.1. Energy requirement

The energy consumed during the PC pyrolysis experiments was estimated from the differences in lower heating values (LHV) of the reactants (PC and CO₂) and the pyrolytic products (syngas, biocrude, and biochar). The LHVs for PC (LHV_{PC}), biocrude ($LHV_{Biocrude}$), and biochar ($LHV_{Biochar}$) were calculated using the Dulong equation ($LHV (\text{MJ kg}^{-1}) = 33.8C + 122.3 \times (\text{H-O}/8) + 9.4S$) (Hosokai et al., 2016). The elemental compositions of the biocrude and biochar were determined using an elemental analyzer (Table 2). The LHV of syngas (LHV_{Syngas}) was calculated based on its composition and the corresponding LHVs of its components: $120.09 \text{ MJ H}_2 \text{ kg}^{-1}$, $10.16 \text{ MJ CO kg}^{-1}$, $49.85 \text{ MJ CH}_4 \text{ kg}^{-1}$, $47.21 \text{ MJ C}_2\text{H}_6 \text{ kg}^{-1}$, $47.74 \text{ MJ C}_2\text{H}_4 \text{ kg}^{-1}$, and $48.47 \text{ MJ C}_2\text{H}_2 \text{ kg}^{-1}$. Additionally, the LHV of CO₂ ($LHV_{CO_2 \text{ at } 600^{\circ}\text{C}}$), used as a feedstock in the pyrolysis test, was determined based on its enthalpy at 600°C ($48.91 \text{ kJ mol}^{-1}$) according to the National Institute of Standards and Technology (NIST) chemistry webbook (Linstrom and Mallard, 2001). The energy requirement for pyrolysis ($LHV_{Pyrolysis}$) was estimated by subtracting the LHV of the reactants ($LHV_{Reactants}$) from that of the products ($LHV_{Products}$), as described in Equations 3–5:

$$LHV_{Reactants} = LHV_{PC} + LHV_{CO_2 \text{ at } 600^{\circ}\text{C}} \quad (3)$$

$$LHV_{Products} = LHV_{Syngas} + LHV_{Biocrude} + LHV_{Biochar} \quad (4)$$

$$LHV_{Pyrolysis} = LHV_{Products} - LHV_{Reactants} \quad (5)$$

2.5.2. Net CO₂ emissions

To assess the impact of CO₂ on the pyrolysis process, all energy-related parameters (including the energy required for pyrolysis and the LHVs of syngas, biocrude, and biochar) were converted into CO₂ emissions. In detail, the energy requirement was considered a positive value when CO₂ is generated from the process, and a negative value when CO₂ is consumed by LHVs. Net CO₂ emissions were calculated by the summation of all CO₂ emission results. For this purpose, the amount of CO₂ produced per unit of heat during diesel combustion ($73.9 \text{ g CO}_2 \text{ MJ}^{-1}$) was used (Pedrozo et al., 2018). The CO₂ adsorption capacity of biochar corresponds to the amount of CO₂ reduced by the biochar. When CO₂ was used as the reactive feedstock, its utilization in the pyrolysis process was estimated by simplifying the reaction pathway between CO₂ and volatile compounds (VCs) as follows: initial VCs ($C_xH_yO_z$) + CO₂ $\leftrightarrow$ reacted VCs ($C_{x-1}H_yO_z$) + 2CO. The variation in CO concentrations

Table 2
Elemental composition of biocrude and biochar derived from the three pyrolysis setups under N₂ and CO₂ atmospheres.

Sample	Pyrolysis setup	C [wt. %]	H [wt. %]	O ^a [wt. %]	N [wt. %]	S [wt. %]	LHV [MJ kg ⁻¹]
Biocrude in N ₂	Single-step	63.66	6.96	29.35	0.03	—	11.11
Biocrude in CO ₂	Single-step	64.02	6.96	29.00	0.03	—	10.98
Biocrude in N ₂	Multi-step	68.44	6.94	24.62	—	—	9.86
Biocrude in CO ₂	Multi-step	67.96	6.94	25.10	—	—	9.58
Biocrude in N ₂	Catalytic	72.97	7.05	19.98	—	—	6.02
Biocrude in CO ₂	Catalytic	73.47	7.03	19.50	—	—	4.63
Biochar in N ₂	Single-step	84.87	2.21	12.92	—	—	—
Biochar in CO ₂	Single-step	84.98	1.73	13.29	—	—	—

^a oxygen content is calculated by following equation: difference (wt.%) = 100-C-H-N-S.

between N_2 and CO_2 environments was then used to quantify CO_2 utilization. In this calculation, the CO increase induced by the reverse water–gas–shift reaction (rWGS), detailed as ' $H_2 + CO_2 \leftrightarrow CO + H_2O$ ', with a conversion efficiency of 78 ± 0.2 wt% (Jung et al., 2020), was additionally considered.

3. Results and discussion

3.1. Thermal degradation behaviors of PC

Pine trees have been widely used in large-scale plantations because of their short logging cycles (Labelle et al., 2017). However, this practice generates a substantial amount of residue, including leaves, branches, cones, and bark, which accounts for approximately 40 wt% of the harvested pine (Pena-Vergara et al., 2022). These residues contain varying amounts of hemicellulose, cellulose, and lignin, which can affect their thermolytic behavior. Therefore, it is crucial to understand the thermal degradation patterns of lignocellulosic residues before conducting pyrolysis tests. To achieve this, TGA of PC was performed under N_2 atmosphere. The mass change (thermogram, TG) and differential TG (DTG) curves of PC are shown in Fig. 1a. In addition, the thermolytic degradation of hemicellulose, cellulose, and lignin was also assessed in N_2 environment for comparison (Fig. 1b).

The thermal degradation of PC can be divided into three distinct phases, separately driven by evaporation (40 – 167 °C), volatilization (167 – 395 °C), and carbonization (395 – 900 °C). At temperatures ≤ 167 °C, a minor mass decrease of ~ 1.9 wt% was observed, which was attributed to the evaporation of residual moisture from the PC despite the pretreatment for drying. At temperatures ≥ 167 °C, a significant

mass reduction of 53.8 wt% was observed up to 395 °C, as characterized by a DTG shoulder at 284 °C and a DTG peak at 348 °C. Subsequently, the degradation was significantly retarded, as evidenced by the suppressed DTG peaks in the temperature range of 395 – 900 °C. These thermolytic patterns resembled the typical degradation patterns observed for lignocellulosic biomass. As the reaction temperature increased, dehydrogenation and deoxygenation occurred concurrently, leading to an enhanced level of carbonization and facilitation of biochar formation.

To further investigate this phenomenon, TGA was performed on hemicellulose, cellulose, and lignin (Fig. 1b). The pyrolytic patterns of these components vary with temperature, reflecting their distinct polymeric structures. Specifically, hemicellulose was degraded at 180 – 350 °C, exhibiting a relatively lower thermal stability than the other components. This broad range of degradation temperatures is attributed to the diverse sugar molecules, such as xylan, mannan, and glucan, which are present in hemicellulose (Brunner, 2014). In contrast, cellulose undergoes pyrolysis in a narrower temperature range of 280 – 360 °C, which is ascribed to the thermal dissociation of the chemical bonds in its glucose-based backbone (Cho et al., 2021). Lignin is a complex molecule primarily composed of phenolic compounds (Jung et al., 2015) and has superior thermal stability compared to that of (hemi)cellulose. The propensity of phenolic compounds to recombine during thermal degradation favors biochar formation from lignin (Galmardini et al., 2013). A comparison of the TGA results for the lignocellulosic components and PC revealed the presence of both (hemi) cellulose and lignin. Notably, the TGA profile of PC closely resembles that of lignin, suggesting that lignin is the primary component of PC.

The thermal decomposition trends of PC under N_2 and CO_2 are

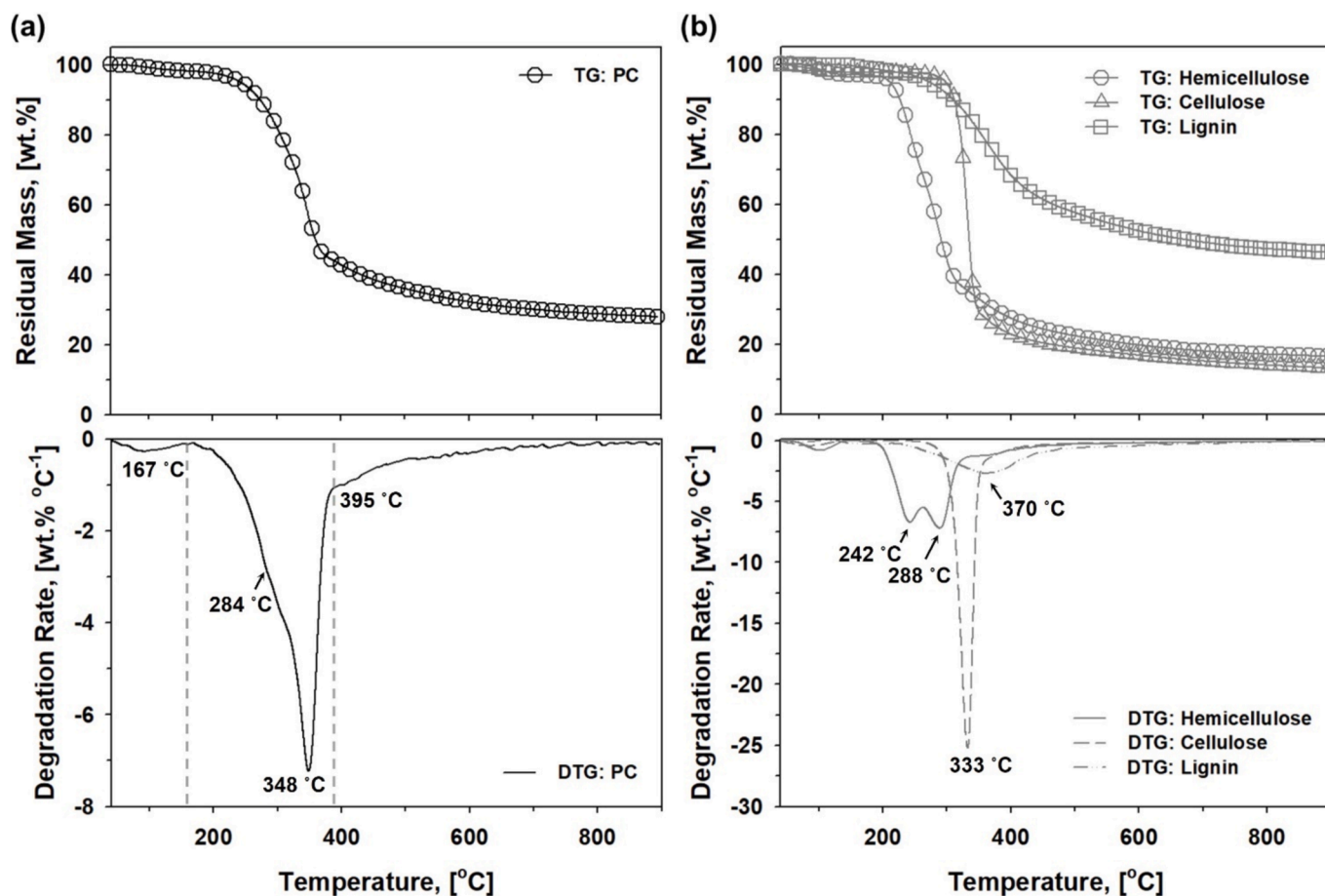


Fig. 1. Thermogravimetric analysis results, mass variation (TG) and degradation rate (DTG), of (a) PC and (b) three lignocellulosic constituents (hemicellulose, cellulose, and lignin) in N_2 atmosphere.

shown in Fig. 2a. At temperatures ≤ 725 °C, the mass change in PC exhibited similar patterns under both N_2 and CO_2 environments, indicating the absence of significant heterogeneous interactions between PC and CO_2 . However, at temperatures ≥ 725 °C, thermolytic behavior was significantly altered in the presence of CO_2 . At 900 °C, the residual mass of PC was 27.9 and 16.8 wt% under N_2 and CO_2 conditions, respectively. The additional mass reduction under CO_2 is likely attributed to its interaction with residual carbon, a process known as the Boudouard reaction: $C_{(s)} + CO_{2(g)} \leftrightarrow 2CO_{(g)}$ (Zhang et al., 2024). While the Boudouard reaction is typically activated at 700 °C (Elleuch et al., 2015), its occurrence at a higher temperature of 725 °C signifies that its reaction kinetics are relatively slow.

To evaluate the extent of the Boudouard reaction, PC was oxidized in air (Fig. 2b). The final residual mass was 2 wt%, which closely corresponds to the ash content (2.1 wt%) determined from the elemental analysis of PC. The substantial difference from the final mass (16.8 wt%) obtained in the presence of CO_2 implies that a significant amount of carbon remained unreacted. This suggests that the Boudouard reaction, induced by CO_2 , occurs at a relatively slow rate. Despite the identified functionality of CO_2 , which expedited the thermal breakdown of PC through heterogeneous reactions, the TGA results did not provide conclusive information about its homogeneous reaction with PC-derived VCs. This limitation of TGA is due to its inability to capture the mass changes associated with gaseous-phase reactions (Chen et al., 2018; Moseson et al., 2020). Therefore, further investigation is necessary to fully understand the homogenous reactions, as the reaction kinetics of CO_2 are likely to be much faster in gaseous-phase reactions (Lahijani et al., 2015).

3.2. Influence of CO_2 on three pyrolytic products

To investigate the effectiveness of CO_2 in gaseous-phase reactions, PC pyrolysis was conducted under N_2 and CO_2 conditions. PC was

pyrolyzed over a temperature range of 100–700 °C at a heating rate of 10 °C min^{-1} , and the resulting pyrolytic products from both conditions were identified and quantified. Syngas (H_2 and CO) and C_{1-2} hydrocarbons (CH_4 , C_2H_6 , C_2H_4 , and C_2H_2), were quantified in real-time. The concentration profiles and total gas production are shown in Fig. 3. Given the negligible production of C_{1-2} hydrocarbons, their concentration profiles were combined.

Under N_2 environment, H_2 generation began at temperatures ≥ 400 °C, which is likely attributed to thermally induced dehydrogenation (Fig. 3) (Kwon et al., 2024). This is consistent with dehydrogenation observed at ≥ 395 °C, as shown in Fig. 1a. The H_2 concentration increased with increasing temperature, reaching a maximum concentration of 1.9 mol% at 700 °C. The production of CO commenced at a temperature of ≥ 220 °C, with a peak concentration of 2.3 mol% at 340 °C, likely originating from the deoxygenation of oxygen-containing functional groups in the (hemi)cellulose-based backbone (Chen et al., 2022). The concentrations of C_{1-2} hydrocarbons peaked at 490 °C (0.6 mol%), indicating the dissociation of methoxy and methyl groups associated with (hemi)cellulose and lignin (Kim et al., 2022).

Under CO_2 environment, the H_2 and CO_2 evolution patterns at ≤ 550 °C were approximately comparable to those observed under N_2 . This indicates that the effect of CO_2 was negligible at lower temperatures. However, gas evolution changed significantly in the 550–700 °C range. There was a marked increase in CO (8.5-fold) compared to that under N_2 at 700 °C. Interestingly, a notable decrease in H_2 production was observed within the same temperature range. Considering that the H_2 decrease coincided with the production of CO , rWGS may have played an important role in enhancing CO production during pyrolysis under CO_2 environment. Moreover, CO production (2.27 mmol) exceeded H_2 reduction (3.97 mmol). Surplus CO cannot be fully accounted for without considering additional inputs from C and/or O sources. Given that the Boudouard reaction for PC was activated at 725 °C and was therefore not responsible for the observed CO enhancement, it is

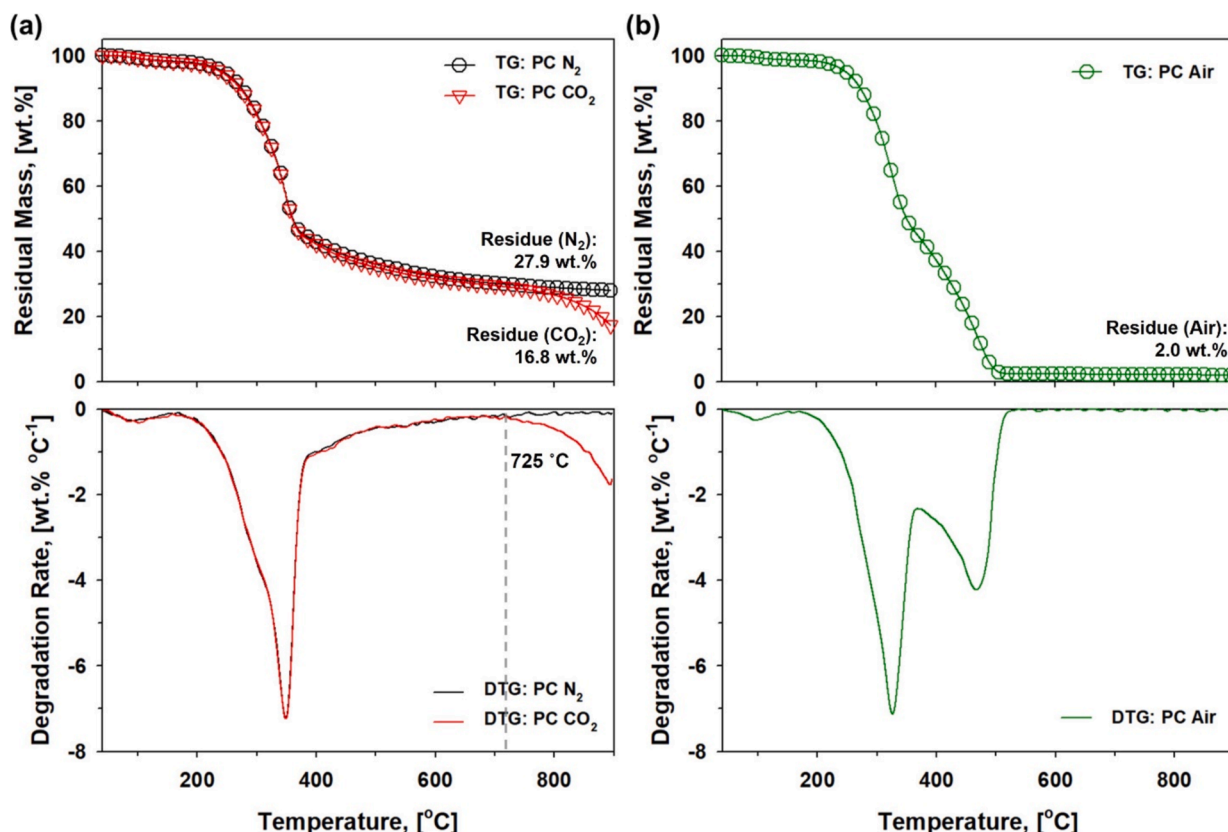


Fig. 2. Thermogravimetric analysis results, mass variation (TG) and degradation rate (DTG) of PC in (a) N_2 and CO_2 , and (b) air atmospheres.

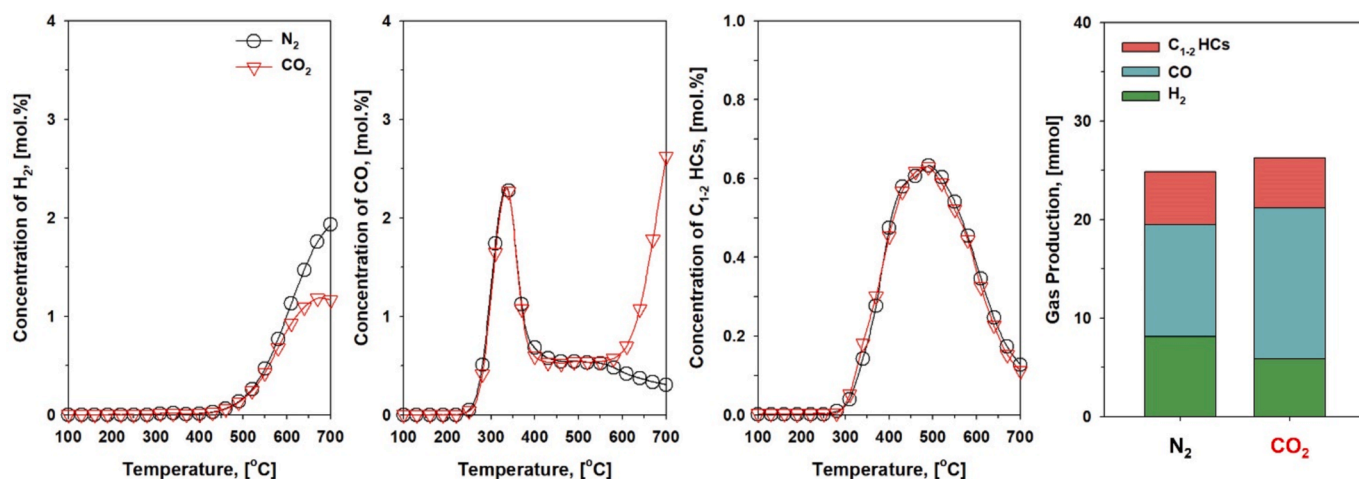


Fig. 3. Evolution patterns of syngas (H₂ and CO) and gaseous hydrocarbons (CH₄ and C₂ hydrocarbons) and their total production from the conventional pyrolysis of PC in N₂ and CO₂ environments.

hypothesized that CO₂ interacts with the volatilized carbon derived from PC. This interaction likely occurred via a gaseous-phase homogeneous reaction, which further enhanced the increase in CO.

Qualitative analyses were performed to assess the effects of CO₂ on the composition of the biocrude under N₂ and CO₂ environments. The GC–MS results for the biocrude are presented in Fig. 4a, and the identified constituents are summarized in Table S2. These chemicals are categorized based on their functional linkages, including carbohydrates, ketones, furanics, phenolics, and polycyclic aromatic hydrocarbons (PAHs) (Qu et al., 2011; Zhou et al., 2014). The cumulative peak areas of the grouped chemicals are shown in Fig. 4b.

The chemical composition of biocrude is closely associated with the relative content of lignocellulosic components in the PC. As shown in Fig. 4b, the biocrude predominantly contained phenolic compounds. Their dominance is likely attributable to the high content of lignin in PC because lignin has a phenol-based three-dimensional framework (Li et al., 2023). The presence of benzene rings in phenols promotes aromaticity, favoring the formation of PAHs in biocrude (Sharma and Hajaligol, 2003). In addition, biocrude contained carbohydrates,

ketones, and furanics, the formation of which is strongly influenced by the thermal decomposition of holocellulose, a mixture of hemicellulose and cellulose (Zhao et al., 2017). The chromatograms of the biocrudes produced under N₂ and CO₂ were notably comparable, suggesting that CO₂ did not significantly affect the chemical species distribution. However, overall biocrude production was reduced under CO₂ environment. This indicated that CO₂ interacted with PC-derived VCs and promoted their conversion into CO. The reduction in individual chemical constituents varied, with the most significant decrease observed in carbohydrates (18.23 %), followed by a 6.47 % reduction in phenolic compounds.

After the pyrolysis experiment, the biochars derived from PC under N₂ and CO₂ environments were retrieved and labeled as PCBN₂ and PCBCO₂, respectively. Fig. 5a displays the surface characteristics of the biochars obtained by SEM. The surface morphology of PCBCO₂ is more uneven than that of PCBN₂. BET/Barrett-Joyner-Halenda (BJH) analysis was employed to examine the texture characteristics of both biochars. The specific surface area of PCBCO₂ (90.3 m² g⁻¹) was 25.8 times greater than that of PCBN₂. As illustrated in Fig. 5b, the adsorption

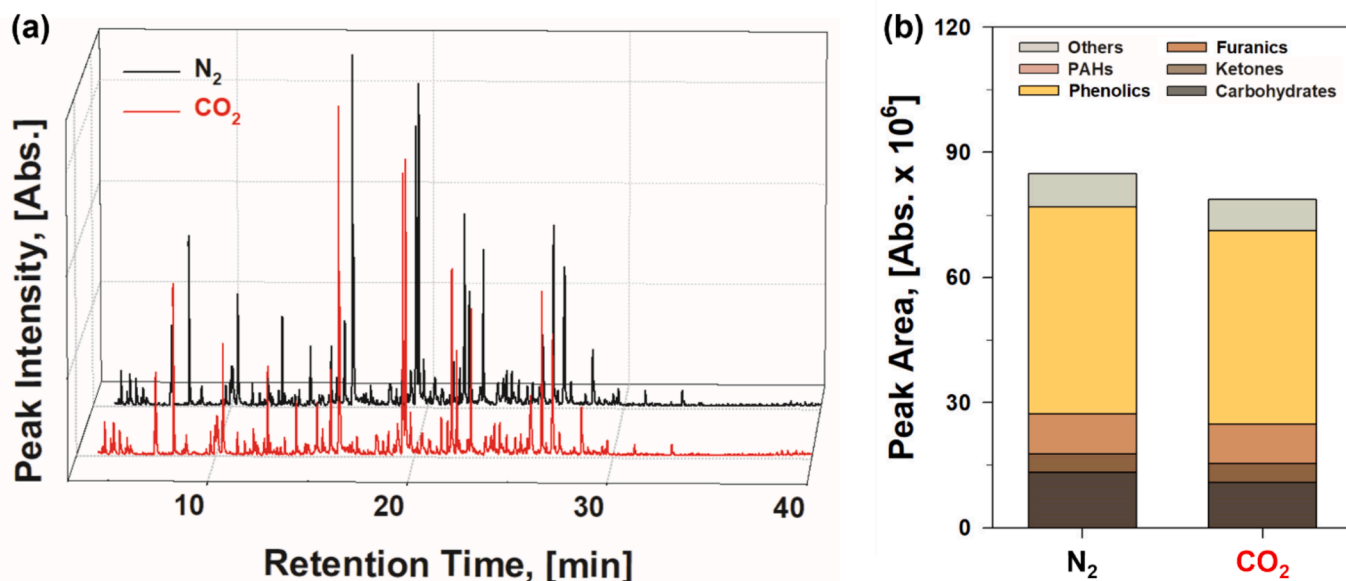


Fig. 4. (a) GC/MS chromatogram results and (b) chemical composition of qualified chemicals in the biocrude produced from PC pyrolysis in N₂ and CO₂ environments.

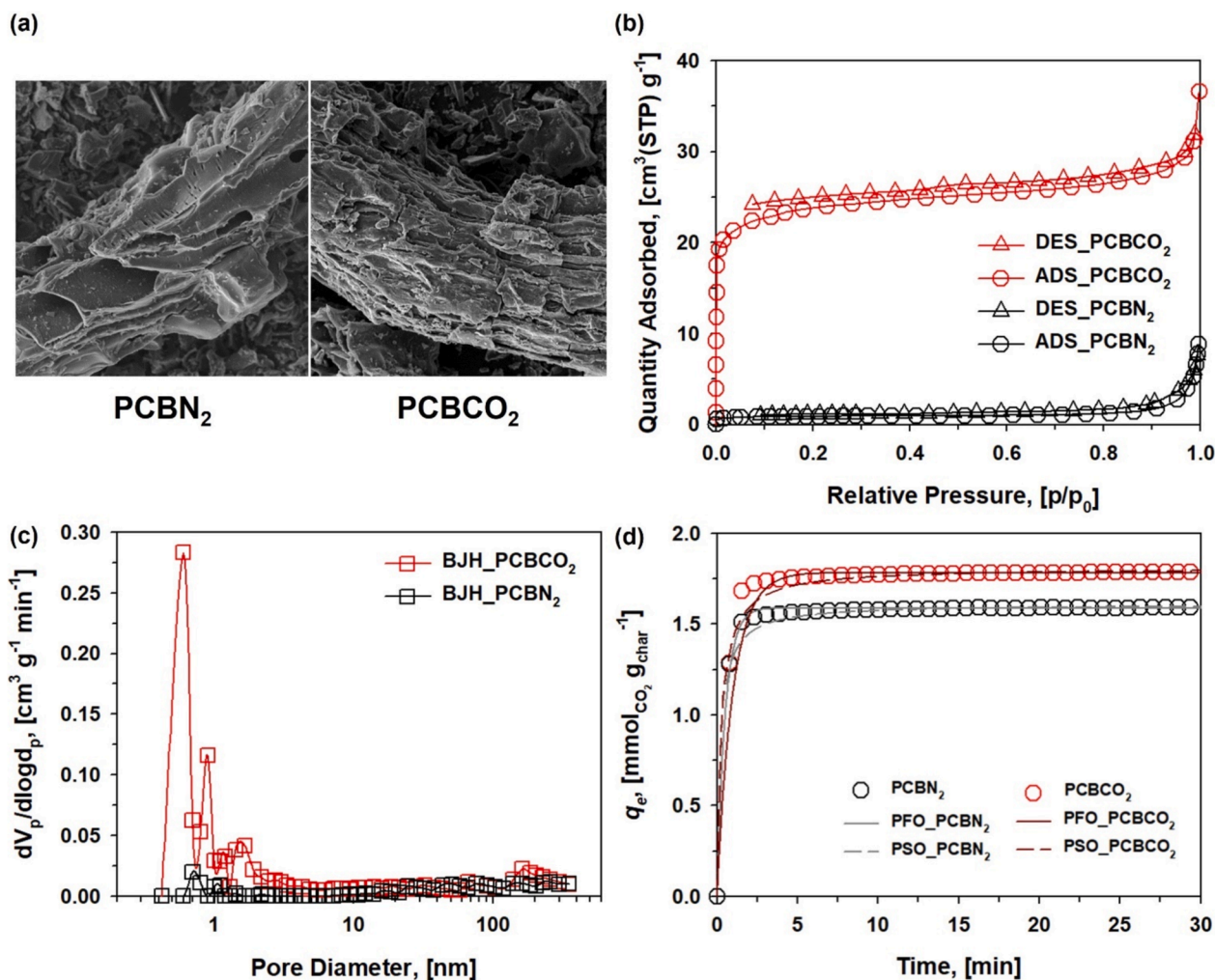


Fig. 5. (a) Surface images (visualized by SEM with a 1000-fold magnification), (b) N₂-physorption profiles (adsorption (ADS) and desorption (DES) curves), (c) pore size distributions (provided from BJH test), and (d) CO₂ adsorption capacities (q_e) and pseudo-first-order (PFO) and pseudo-second-order (PSO) fittings of PCBN₂ and PCBCO₂ produced from PC pyrolysis in N₂ and CO₂ environments.

(ADS)/desorption (DES) curves of PCBN₂ show a Type III isotherm, indicating the development of macro- and mesopores (Burbano et al., 2023). In contrast, PCBCO₂ displayed Type I and III isotherms in the ADS/DES curves, indicating the predominant development of

micropores. This is further confirmed by the pore-size distribution shown in Fig. 5c (Zhang et al., 2023). The mean pore diameters of PCBN₂ and PCBCO₂ were 5.6 and 1.1 nm with the total pore volumes of 0.01 and 0.05 cm³ g⁻¹, respectively. The enhanced porosity of PCBCO₂

Table 3

Comparison of the biochar performance for the CO₂ adsorption.

Biochar production Feedstock	Biochar characteristics				CO ₂ adsorption capacity (q_e), [mmol g ⁻¹]	Kinetic rate, [min ⁻¹]	Ref.
	Pyrolysis conditions	Specific surface area, [m ² g ⁻¹]	Total pore volume, [cm ³ g ⁻¹]	Mean pore diameter, [nm]			
pine cone	700 °C, N ₂ , 0.5 h	3.5	0.01	5.6	1.5	1.9	This study
pine cone	700 °C, CO ₂ , 0.5 h	90.3	0.05	1.1	1.8	1.0	This study
wheat straw	600 °C, 3 h	123.2	0.11	1.7	0.8	1.0	(Cao et al., 2022)
walnut shell	600 °C, 3 h	336.7	0.20	1.2	1.0	0.8	(Cao et al., 2022)
peat	600 °C, Ar	237.7	0.18	—	0.9	—	(Taher et al., 2023)
corn stalk	600 °C, 3 h	36.1	0.10	5.6	0.9	0.3	(Zhang et al., 2022)
soybean straw	600 °C, 3 h	109.6	0.06	1.0	0.9	0.4	(Zhang et al., 2022)
sugarcane bagasse	600 °C, N ₂ , 2 h	388.3	—	—	1.7	—	(Creamer et al., 2014)
vine shoots	600 °C, N ₂ , 1 h	244.0	0.07	—	2.1	—	(Manyà et al., 2018)
olive stones	800 °C, N ₂ , 2 h	774.0	0.30	0.6	2.7	—	(González et al., 2013)
almond shell	800 °C, N ₂ , 2 h	721.0	0.30	0.7	2.5	—	(González et al., 2013)

indicates its superior potential as an effective environmental adsorbent (Kim et al., 2019; Premchand et al., 2023).

To evaluate the utility of the PC-derived biochars, their CO₂ adsorption capacities were measured using TGA. The CO₂ adsorption profiles were modeled using two different fitting approaches: pseudo-first-order (PFO) and pseudo-second-order (PSO) equations. The PFO and PSO fitting results are presented in Fig. 5d and Table 3, respectively. The PFO model provided a more accurate description of CO₂ adsorption kinetics on the two biochars (PCBN₂ and PCBCO₂) compared to the PSO model, indicating that the adsorption of CO₂ is predominantly influenced by physisorption mechanisms, which are closely related to Van der Waals forces (Wjihi et al., 2021; Yahia et al., 2024). The notable increase in the specific surface area of PCBCO₂ led to a 1.2-fold increase in the CO₂ adsorption capacity compared with that of PCBN₂. Compared to other biochars, PCBCO₂ demonstrated competitive CO₂ adsorption capacity (Table 3), confirming that pyrolysis in CO₂ environment enhanced the effectiveness of biochar as an adsorbent.

3.3. Enhanced functionality of CO₂ from catalytic pyrolysis

Despite the mechanistic function of CO₂ observed during PC pyrolysis, its activation occurred only at temperatures of ≥ 550 °C. Given that the major volatilization of PC occurs at temperatures ≤ 550 °C, the absence of noticeable gaseous phase interactions with CO₂ at this lower temperature is likely attributable to the slow rate of reaction kinetics. To address this issue, a modified pyrolysis setup was employed by adding

an additional heat block maintained at a constant temperature of 600 °C. The specified temperature was determined within the temperature range of 550–700 °C where the additional formation of CO occurred through CO₂-mediated gas phase reactions without occurrence of the Boudouard reaction, as confirmed in Fig. 3. This setup, referred to as ‘multi-step pyrolysis’, was designed for a more accurate simulation and evaluation of the interactions between CO₂ and PC. The concentrations of the key gaseous products were monitored and profiled as a function of the reaction temperature, as shown in Fig. 6a.

The overall evolution patterns from the multi-step setup under N₂ environment were approximately comparable to those from the single-step pyrolysis (Figs. 3 and 6a). However, there was a slight increase in gas concentrations at 200–400 °C, with H₂ at 0.3 mol%, CO at 3.8 mol%, and C_{1–2} hydrocarbons at 1.0 mol%. These enhancements were attributed to further thermal cracking of the VCs that evolved from the PC. Despite these increases, the overall gas productivity under N₂ and CO₂ conditions was comparable. This suggests that the additional heat block of 600 °C did not effectively improve the reaction kinetics of CO₂.

To enhance gaseous-phase interactions, a Ni catalyst (4.8 wt% Ni/Al₂O₃) was placed in the center of the second heat block, termed ‘catalytic pyrolysis’. A Ni catalyst was selected because of its capability to activate dehydrogenation and deoxygenation (Wang et al., 2020; Yang et al., 2023). When Ni-based catalysts are used for commercial applications, their typical Ni loading is in the range of 5–40 wt%. However, using lower Ni loading helps enhance the economic feasibility of catalytic process. As such, only 4.8 wt% Ni/Al₂O₃ catalyst was employed in

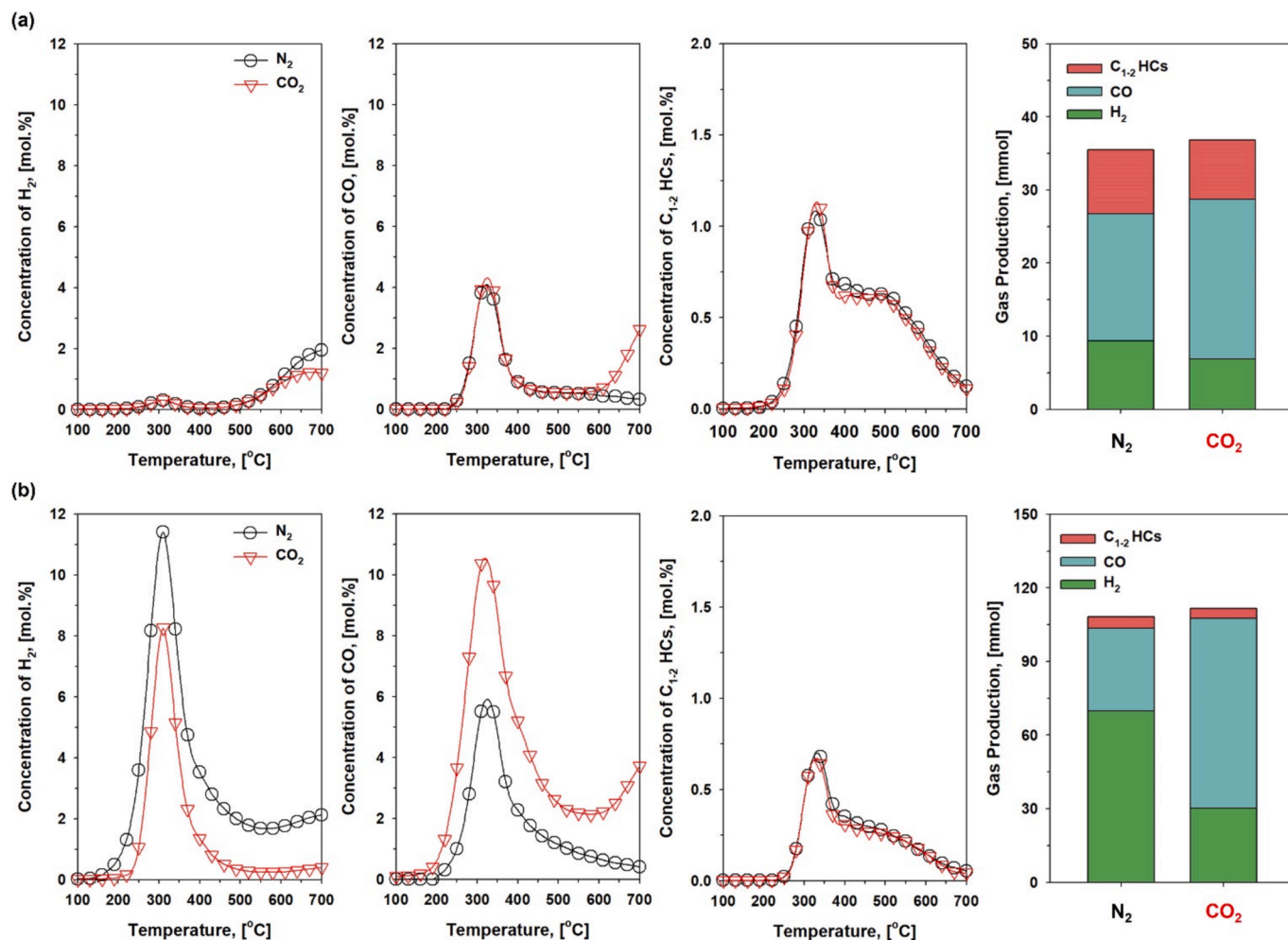


Fig. 6. Evolution patterns of syngas (H₂ and CO) and gaseous hydrocarbons (CH₄ and C_{1–2} hydrocarbon mixture) and their total production produced from (a) multi-step and (b) catalytic pyrolysis of PC from N₂/CO₂ atmospheres.

this study. The evolution patterns of the key gaseous products from catalytic pyrolysis are shown in Fig. 6b. The overall evolution trends from the catalytic pyrolysis under N_2 showed a notable enhancement compared to the multi-step setup. The production of H_2 peaked at $310^\circ C$ with a maximum concentration of 11.4 mol%, leading to a ~ 38 -fold increase. The significant enhancement in H_2 production was attributed to the catalytic activation of the C-H bonds by Ni. The preferential activation of C-H bonds led to reduced formation of C_{1-2} hydrocarbons. The Ni catalyst also promoted CO production, achieving 5.50 mol% at $310^\circ C$, which exceeds the CO production from the multi-step pyrolysis by 1.41 times.

In contrast to catalytic pyrolysis in N_2 atmosphere, the gas evolution under CO_2 environment was considerably altered across the entire temperature range (100 – $700^\circ C$). A notable increase in CO production was observed, achieving 10.4 mol% at $310^\circ C$, which is 1.9 times higher than that under N_2 condition. Conversely, the peak concentration of H_2 under CO_2 environment was 8.3 mol%, showing a 1.4-fold decrease compared to that under N_2 environment. The concentration profiles of C_{1-2} hydrocarbons were comparable under both conditions. This indicates that CO_2 played a significant role in altering H_2 and CO formation. In this context, the rWGS played an important role in the production of CO by consuming H_2 and CO_2 . Excess CO production compared to H_2 reduction further supports the mechanical function of CO_2 in enhancing CO formation during catalytic pyrolysis.

To further investigate the effect of CO_2 , the peak intensities of the biocrude constituents generated from all test setups for pyrolysis were compared under N_2 and CO_2 environments. The results of the qualitative analyses of the biocrudes are summarized in Tables S2–S4. The chemicals were categorized according to their functional groups, and their peak area summations are shown in Fig. 7a. The data revealed that the production of grouped chemicals decreased stepwise when an additional heat block and Ni catalyst were added to the pyrolysis setup. The use of CO_2 further suppressed the peak area of these chemicals. These results are consistent with the CO_2 's reactive functionality, which promotes the transformation of PC-derived VCs into CO. The overall mass balances of the resulting three products are shown in Fig. 7b. Despite a similar mass of biochar across all pyrolysis setups, the interaction of CO_2 with the VCs led to a reduction in biocrude production and a simultaneous increase in syngas production. The effectiveness of CO_2 was notably enhanced when

combined with the Ni catalyst, demonstrating its superior role in PC pyrolysis. Nevertheless, since this study utilized the Ni catalyst with a specific nickel loading (4.8 wt% Ni/ Al_2O_3), preparing Ni catalysts with varying loadings and testing them in catalytic pyrolysis could help further optimize the CO_2 's functionality. Thus, these follow-up studies should be conducted in the near future.

3.4. Evaluating sustainability of the CO_2 -mediated pyrolysis

To assess the environmental feasibility of the proposed pyrolysis system, it is crucial to investigate the energy required for pyrolysis tests and the net CO_2 emissions. A detailed process flow diagram to estimate these two variables is shown in Fig. 8a. Considering that pyrolysis is an endothermic reaction involving the thermal breakdown of chemical bonds in carbonaceous materials (Somers et al., 2018), the energy required for pyrolysis is a critical factor in evaluating the sustainability of the proposed system. The energy requirements were significantly influenced by the process parameters. To assess this, the LHV of the reactants (PC and CO_2) and three pyrolytic products (syngas, biocrude, and biochar) were determined assuming complete combustion. The energy requirements for pyrolysis were determined by calculating the differences between the LHVs of the products and reactants. The resulting energy balances are shown in Fig. 8b and Table S5.

Modifying the pyrolysis test configuration under N_2 condition did not affect the LHV of the feedstock because it only considered the chemical potential of PC. However, the LHV value under the CO_2 condition showed a slight increase. Such an effect of CO_2 on elevating feedstock's LHV was attributed to its participation through the gaseous-phase interaction. Although the LHVs for biochar were similar across all the pyrolysis experiments, those for syngas and biocrude were different. Overall, the LHVs of the three pyrolytic products were significantly higher than those of the feedstock. Thus, the energy required for each pyrolysis test was confirmed.

Subsequently, the energy requirements for the pyrolysis tests were utilized to calculate CO_2 emissions by converting the required energy into the amount of CO_2 emitted when diesel fuel is burned. Combustion of diesel, one of the primary fuels in various industries and transportation sectors, accounts for a significant portion of total anthropogenic CO_2 emissions. Therefore, the use of diesel as a reference fuel for

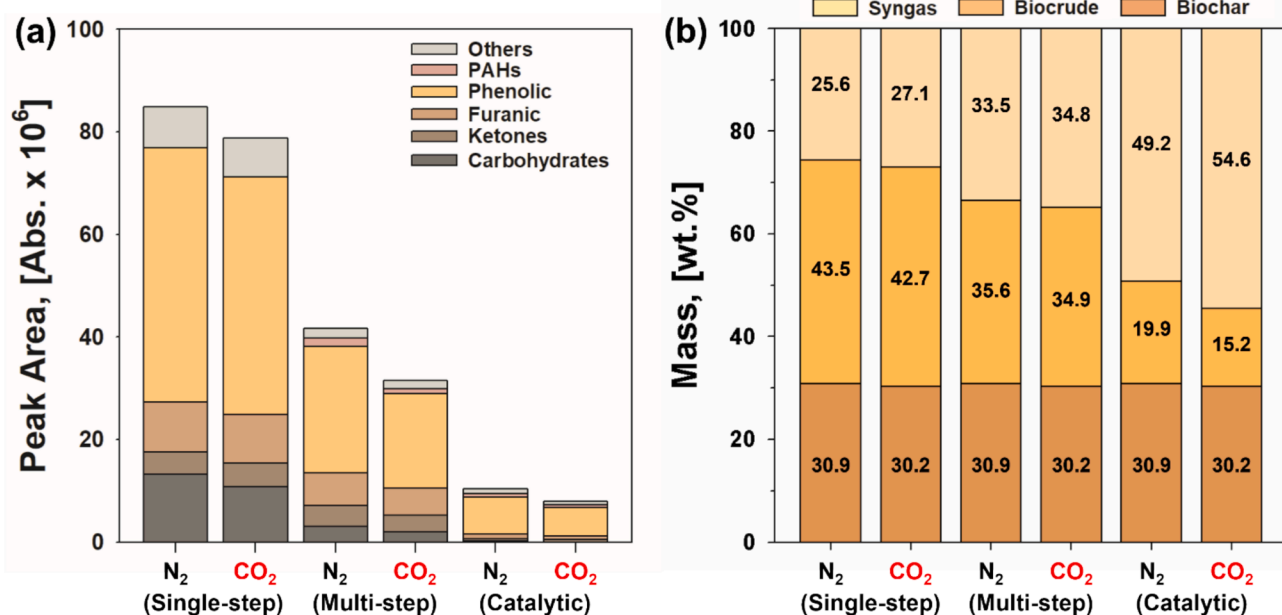


Fig. 7. (a) Total peak area of qualified constituents of biocrude and (b) overall mass balances of syngas, biocrude, and biochar from all test setups for pyrolysis of PC in N_2 and CO_2 environments.

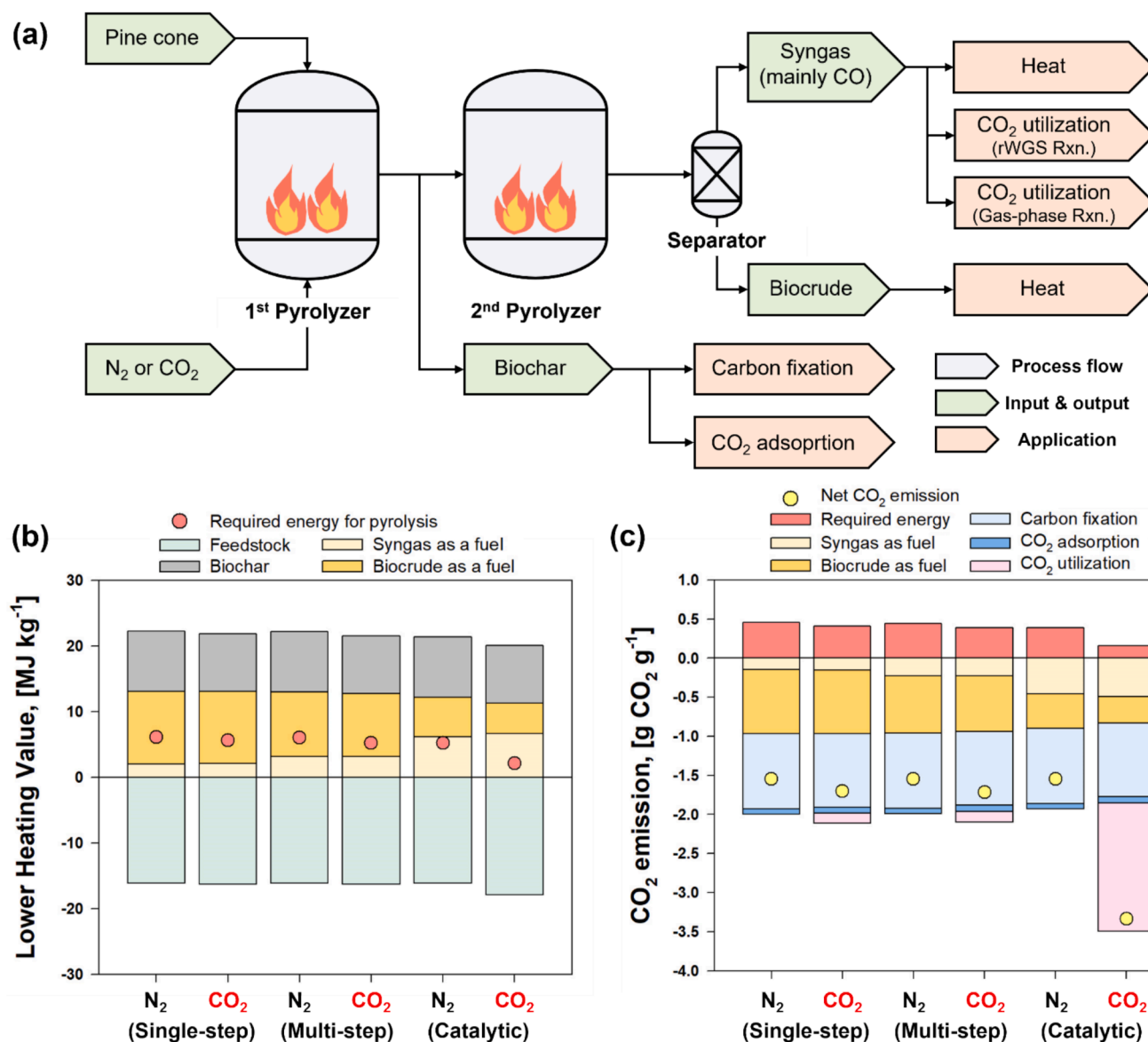


Fig. 8. (a) Process flow diagram, (b) energy requirement for pyrolysis, and (c) net CO₂ emissions obtained from three pyrolysis tests for PC under N₂ and CO₂ environments.

calculating CO₂ emissions could hold a potential to reflect realistic scenarios. Given the inherent carbon neutrality of PC, using the pyrolytic products, syngas and biocrude, as energy sources was considered to reduce CO₂ emissions. During syngas production, CO₂ contributes to CO formation via two routes, rWGS and gaseous-phase reactions, which exemplifies the utilization of CO₂. In addition, biochar reduces CO₂ emissions through carbon fixation. The carbon content in the biomass was calculated as CO₂-equivalent, and the CO₂ adsorption results (from Fig. 5d) were also included. Based on this rationale, the net CO₂ emissions from the proposed pyrolysis system were estimated and presented in Fig. 8c and Table S6. Catalytic pyrolysis of PC in the presence of CO₂ demonstrated superior performance in reducing CO₂ emissions, achieving 3.34 g CO₂ per gram of PC. This study confirms that introducing CO₂ into a pyrolysis system offers significant environmental advantages and enhances the sustainability of the biomass conversion process.

While the CO₂'s functionality in enhancing the sustainability of biomass conversion process was identified, laboratory-scale experiments face technical challenges in achieving similar results in scaled-up applications. To address this challenge, numerous studies have

conducted TGA tests on lignocellulose under various heating rates, quantifying the resulting pyrolytic products (Piazza et al., 2024; Quan et al., 2016; Xiao et al., 2020). These analyses provide the kinetic parameters, such as activation energy, which are critical for larger reactor design. Applying these values in the computational tools, such as Aspen Plus, can facilitate the large-scale feasibility of the proposed system (Peters et al., 2017). However, despite these kinetic values, scaling up reactor designs may pose several technical challenges, primarily resulting from heat and mass transfer limitations. As reactor size increases, the heat gradient from a heat source to a center becomes more pronounced, leading to the retarded thermal transfer. This uneven heat distribution complicates the production of consistent pyrolytic products. The poor heat transfer adversely influences the generation of VCs from thermal cracking of lignocellulose, retarding their mass transfer into CO₂. This delay in CO₂'s reactivity may hinder the efficiency of the pyrolysis process. Therefore, conducting thermal-fluid analysis using computational fluid dynamics (CFD) models could help optimize reaction parameters and improve the environmental feasibility of CO₂-mediated pyrolysis for the industrial-scale implementation (Lu et al., 2022).

4. Conclusions

This study emphasizes the development of a pyrolysis method for the comprehensive utilization of lignocellulosic biomass (representatively PC). The incorporation of CO₂ into the pyrolysis process is advantageous because of its effectiveness as a reactive feedstock. The initial TGA of the PC confirmed that lignin was the primary component. Laboratory-scale pyrolysis tests under N₂ and CO₂ atmospheres revealed that CO₂ facilitated compositional changes in the pyrolytic products of syngas, bio-crude, and biochar. The interaction of CO₂ with PC-derived VCs decreased biocrude and increased the gaseous products, particularly CO. This homogeneous reaction also affected the biochar's surface characteristics, notably increasing its specific surface area and creating micropores that improved its CO₂ adsorption capacity. The efficacy of CO₂ was further demonstrated through modifications to the experimental setup, including multi-step and catalytic pyrolysis, which further enhanced CO production. To evaluate the sustainability of the proposed system, energy requirements and CO₂ emissions were calculated for each pyrolysis test. Catalytic pyrolysis under CO₂ environment proved most efficient, offsetting 3.34 g of CO₂ per gram of PC. While CO₂ contributes to the sustainability of biomass conversion process, this study does not completely resolve challenges in its industrial and practical applications. Designing the larger-scale pyrolysis reactor may face critical technical hurdles, such as heat and mass transfer limitations. To address these, the use of computational tools, including Aspen and CFD modelling, is strongly recommended. Therefore, future studies should focus on optimizing process variables and improving environmental feasibility of CO₂-mediated pyrolysis for industrial-scale implementation.

CRediT authorship contribution statement

Kwangsuk Yoon: Writing – original draft, Formal analysis, Data curation. **Taewoo Lee:** Writing – original draft, Validation, Data curation. **Hoyeon Cha:** Validation, Methodology. **Joohyung Lee:** Visualization, Investigation. **Jegeon Lee:** Investigation, Formal analysis. **Hocheol Song:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.biortech.2024.131765>.

Data availability

Data will be made available on request.

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